ECG-Based Indicators of Underlying Cardiac Disease and Arrhythmia Presence

Basic Info:

- **Git Hub Repository:** https://github.com/Tarra242/Data-Wrangling-Group-1-
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Background and Motivation:

Our motivation for selecting this project stems from our shared clinical backgrounds and interest in applying data-driven methods to clinically relevant problems. In clinical settings, electrocardiograms (ECG or EKGs) are routinely used to assess cardiac function and identify arrhythmias, yet interpretation often relies on pattern recognition and clinician experience. We are interested in evaluating how quantitative ECG-derived features can be used to better understand arrhythmia presence and variation across arrhythmia types.

Project Objectives

The primary questions we are trying to answer are these;

- How do ECG-derived features differ between individuals with normal rhythm and those with cardiac arrhythmia?
    - We are aiming to identify which ECG measurements are most strongly associated with the presence of arrhythmia.
- Is there a cumulative effect of ECG abnormalities on arrhythmia presence?
    - We aim to assess whether an increasing burden of ECG abnormalities corresponds to a higher likelihood of arrhythmia.
- Do ECG feature patterns vary across arrhythmia subtypes?
    - We are going to explore whether different arrhythmia classes exhibit distinct ECG profiles or are similar to normal ECGs.
- How much do age and weight affect baseline ECG measurements across different arrhythmia types?
    - We aim to assess whether the patterns identified for classification suffer influence of age and weight

Data (See Works Cited Information Below)

https://archive.ics.uci.edu/dataset/5/arrhythmia

The data used in this project are from the Arrhythmia Data Set hosted by the UCI Machine Learning Repository. It is a publicly available dataset originally compiled by H. Altay Guvenir and colleagues for the purpose of studying ECG-based arrhythmia classification. It includes 452 patient records with 279 ECG-derived features and cardiologist-assigned diagnosis labels. The dataset was downloaded directly from the UCI repository in its original format and loaded into a Jupyter Notebook for initial analysis using Python.

Data Processing

We plan to have moderate to heavy preprocessing to analyze the dataset size (279 features), mixed attribute types, and the reformatting of the raw files. Initial analysis showed that the raw data required careful editing (e.g., loading with header=None to avoid treating the first data row as column headers). We will also check for missing values, inconsistent data types, and any duplicate or redundant information that is created during the first and final walk-throughs.

The cleaning process starts with correctly importing the raw data file with the appropriate setting. We will then rename the key variables (e.g., the diagnosis label column) for visual clarity. Next we will identify '?' placeholders for the missing values and resolve non-numeric values. Initial analysis shows that there might be some columns we will remove given the lack of data or importance to our goal. Missing values will lead to mean/media imputation from specific groups or record exclusion, depending on the interpretation of the group. Overall, we will standardize the continuous features prior to execution. The overall processing will be documented step by step in Jupyter notebooks, including data validation checks (e.g., confirming diagnosis label distribution, verifying variables, and confirming there are no duplicated columns). This will ensure transparency and reproducibility for both the cleaning and execution stages. (See our Group 1 GitHub Repository for our initial analysis of the data.)
https://github.com/Tarra242/Data-Wrangling-Group-1-/tree/main

Design

The main goal of our visualizations is to help us explore and understand the Arrhythmia dataset so we can make good decisions during data cleaning. Because the dataset has many features and missing values in ECG-related variables, we will start by visualizing the patterns of missing data and how well the classes (arrhythmia types) can be separated. We will create visualizations that show where data is missing. These plots will use simple color coding (e.g., one color for "missing" and another for "present") so we can quickly see which features have the most

missing values and whether missingness occurs in certain groups of features. This helps us spot patterns and decide how to handle missing data. Next, we will group ECG features into meaningful physiological categories. For each group, we will show basic summary statistics for the different arrhythmia types. This makes the dataset easier to understand and helps highlight which feature groups differ most across conditions. Alternatively, we could create a design where the different physiological categories or the ECG features could be summarized in a radar/spider plot, so that the interpretation is made easier.

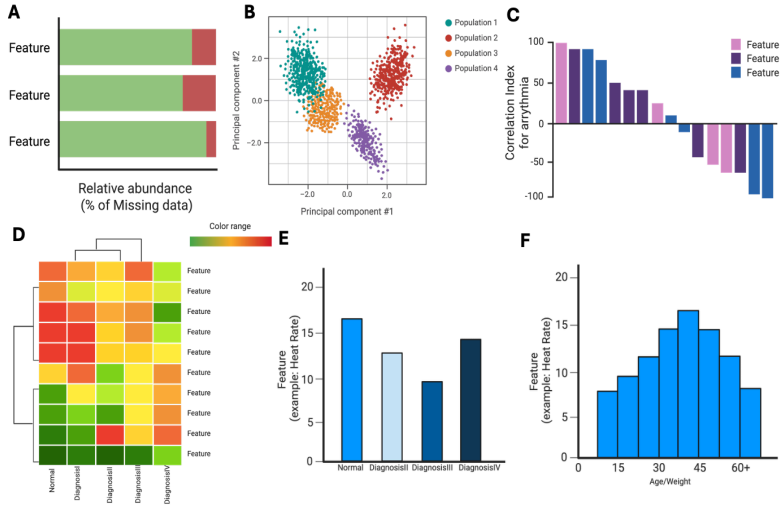

Figure 1. Main visualization strategies. A) Visualization of missing values in features. **B)** Computing features into Principal Components to create clusters related to diagnosis **C)** After determining a correlation index between the features and the different arrythmia diagnosis, we can determine which ones are more important **D)** A heatmap will demonstrate which features display more variation from the normal profile. **E)** For each feature, we can alternatively create a bar plot to show their difference to the normal values for each feature we select. **F)** Distribution of the feature values across age and/or weight.

Must-Have Features

This project will be considered unsuccessful if it cannot support meaningful comparison of ECG patterns across arrhythmia subtypes. As such, the following features are essential: a cleaned and standardized ECG features matrix, an explicit and documented handling of missing data, a clear definition of arrhythmia types, visualizations that enable comparison of ECG patterns across the types, and a reproducible data wrangling pipeline (a clear documentation of the preprocessing steps).

Optional Features

- Interactive or enhanced visualization, e.g., hoverable plots.
- Additional research and ECG labels with subtype classes.
- Summary measures or ECG abnormalities in depth if time permits, but it is not necessary.
- Linear regressions to test our missing value adjustments.

Project Schedule: (Divided by 5 group members–indivdual assignments to be determined.)

By February 15th:

- We will confirm the dataset has been correctly imported into github and add appropriate headers and organize the data into a clear, easy-to-access table.

- We will conduct Initial preprocessing and data cleaning - determining which columns will be used and how we are going to deal with the columns that have missing values (e.g., mean or median).
- We will assess missing values and determine appropriate handling strategies.
- We will Identify which variables we will use for our data visualization.
- We will determine which type of graphs will be used for our data visualization.

By February 28th:

- Data validation check. We will validate key variables (e.g., diagnosis labels, arrhythmia indicators and demographic fields).

By March 2nd: First round of peer reviews.

- Create graphs for our visualization.
- We will determine which type of graphs will be used for our data visualization.
- We will create graphs for analysis and visualization.
- We will create our Linear Regressions.
- We will write out our Read me in github.

Further assessment on roles and schedule will be assigned among team members upon first peer review.

Works Cited

Guvenir, H. A., Acar, B., Demiroz, G., & Cekin, A. (1997). *Arrhythmia data set*. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/5/arrhythmia